



# dxagent

An agentic diagnostic reasoner for acute dyspnoea and chest pain  
how it works, what was measured, and what is wrong with it

177 likelihoods · 39% carrying a citation · 19 real patients · 170 tests

Mohamed Aziz Ben Rejeb · DeepShift AI Summer Internship 2026, Project 1 (Healthcare)

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Not for clinical use. No part of this has been reviewed by a clinician.



# 1 The problem, and the scope decision behind it

A patient arrives short of breath, with chest pain, and nobody yet knows why. Eight conditions could explain it and they compete for the same evidence. Three of them kill people within hours. The task is not to name a disease from a complete record — it is to decide, one question at a time, what to ask next, and when the answer is certain enough to act on.

## The eight candidates

| Cause                        | Why it is in scope                                            | Red flag |
|------------------------------|---------------------------------------------------------------|----------|
| Pulmonary embolism           | Time-critical, commonly missed, published rules (Wells, PERC) | yes      |
| Community-acquired pneumonia | The most common competitor                                    |          |
| Acute coronary syndrome      | Time-critical, published rule (HEART)                         | yes      |
| Acute pulmonary oedema       | The acute decompensation of heart failure                     | yes      |
| COPD exacerbation            | Overlaps heavily on dyspnoea and wheeze                       |          |
| Asthma exacerbation          | Separating it from COPD is a real clinical task               |          |
| Pericarditis                 | Shares pleuritic pain with pulmonary embolism                 |          |
| Panic attack                 | The benign cause that must not be reached by exclusion alone  |          |

## One presentation, not three

The brief suggested three to five presentations. This covers one, deliberately. Hyponatraemia, anaemia and unexplained dyspnoea are separate diagnostic problems with disjoint evidence sets; three of them covered thinly produces a knowledge base too shallow to discriminate *within* any one of them. One presentation with eight genuinely competing causes exercises the same machinery against a differential where the candidates actually fight each other.

### Asymmetric by design

Three causes are time-critical. The gate will not commit to a *more likely* benign diagnosis while one of them still holds meaningful probability.

Missing a panic attack and missing a pulmonary embolism are not the same error.

### 29 findings, 177 cells

Eight diseases against the findings each one describes gives the likelihood grid the rest of this document keeps returning to.

It is simultaneously the engine and the principal weakness.

## The findings it can ask about

| Where it comes from           | n  | Findings                                                                                                                                                                                                       |
|-------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| History — free, instant       | 13 | calf tenderness, dyspnoea at rest, exertional chest pain, fever, leg swelling, orthopnoea, palpitations, pleuritic pain, productive cough, recent immobility, smoking history, sudden onset, wheeze subjective |
| Examination — cheap, one turn | 8  | crackles, ecg pr depression, ecg st changes, friction rub, hypoxia, raised jvp, reduced breath sounds, tachycardia                                                                                             |
| Laboratory — costed, delayed  | 4  | raised bnp, raised d dimer, raised troponin, raised wcc                                                                                                                                                        |
| Imaging — most expensive      | 4  | ctpa filling defect, cxr consolidation, cxr pulmonary oedema, pericardial effusion                                                                                                                             |

The costs are one of the invented families counted in section 5: an ordinal ranking a non-clinician can defend, not a tariff from any health system.

## 2 How it works

Four layers and a loop. The loop runs until the gate commits, escalates, or a budget runs out.

| Layer         | What lives there                                                                                                                                                                       |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Knowledge     | The likelihood tables, Wells / PERC / CURB-65 / HEART with citations, Merck Manual sentences, the findings-and-causes graph, and the provenance record of where every number came from |
| Reasoning     | Naive Bayes in log space with correlation weighting, the information-gain action selector, temperature scaling, conformal prediction and the abstention gate                           |
| Orchestration | A hand-written reference loop and the same loop as a LangGraph state machine, held equivalent by a test that runs both over every case and asserts identical verdicts                  |
| Evaluation    | Ranking, calibration, selective prediction, DDx recall, and the two baselines the brief requires                                                                                       |

### The five steps of one turn

#### 1 Propose

Rank the eight causes on the findings so far. Correlation weighting stops four facets of one clinical picture counting as four independent pieces of evidence.

#### 2 Calibrate

Temperature-scale the posterior. The scaler refuses to fit below 30 labelled cases, and the report now says so rather than printing a default as a finding.

#### 3 Check the gate

Six conditions, all of which must hold before committing.

#### 4 Choose the next step

Highest expected information gain per unit cost, with a red-flag weight and a two-tier rule preferring a decisive test over a marginally cheaper question.

#### 5 Ask

Put it to the patient or the record. A finding the source never recorded costs neither a turn nor money — no test was performed.

### Two engines, held equivalent by a test

The same loop exists twice: a hand-written reference implementation, and a LangGraph state machine of four nodes — propose, calibrate, decide, observe — with the decide node routing back to propose or out to a verdict. Having two is only defensible if they cannot disagree, so a test runs both over every case and asserts identical verdicts, identical predictions and identical evidence sequences. The graph is the version a reviewer can read as a diagram; the loop is the version that runs when LangGraph is not installed.

#### Selector: gain per unit cost

Expected reduction in entropy over the eight causes, divided by what the action costs, with an extra weight when it bears on a time-critical cause.

#### The tie-break that had to be added

Pure gain-per-cost prefers a nearly free question with slight gain over a decisive scan. A two-tier rule takes the decisive action when the cheap one would not move the gate.

## The six conditions that must all hold before it answers

| # | Condition                                          | Why                                                                              |
|---|----------------------------------------------------|----------------------------------------------------------------------------------|
| 1 | Calibrated top-1 probability $\geq 0.65$           | A floor on certainty                                                             |
| 2 | Top-two margin $\geq 0.15$                         | Catches contested posteriors a probability threshold alone waves through         |
| 3 | Top hypothesis grounded in the KB, with a citation | No answer without a source                                                       |
| 4 | <b>No red-flag diagnosis above 0.10</b>            | The asymmetry: will not commit to a likelier benign cause while a killer is live |
| 5 | Proposer disagreement below threshold              | When two proposers run, disagreement is a signal                                 |
| 6 | The evidence is explained by something             | Refuses to answer on findings the KB cannot account for                          |

Failing any one produces an escalation carrying a stated reason and a named unresolved question — not a silent low-confidence answer. Several now read *“Pulmonary embolism exclusion could not be completed — sought but not available: lab:raised\_d\_dimer”*.

## One real case, turn by turn

23-year-old man, one day of right-sided pleuritic chest pain, haemoptysis and fever — case pmc-6129844, a published case report, run through the shipped configuration. Nothing here is scripted: the “why that one” column is the selector’s own reason.

| #  | Asked for             | Why that one                                                | Answer       | Top-1 after                      |
|----|-----------------------|-------------------------------------------------------------|--------------|----------------------------------|
| 1  | raised d dimer        | rule out pulmonary_embolism (3%); discrimination 0.62       | not recorded | acute coronary syndrome 25%      |
| 2  | raised troponin       | rule out acute_coronary_syndrome (25%); discrimination 0.68 | not recorded | acute coronary syndrome 25%      |
| 3  | ctpa filling defect   | rule out pulmonary_embolism (3%); discrimination 0.78       | absent       | acute coronary syndrome 25%      |
| 4  | exertional chest pain | rule out acute_coronary_syndrome (26%); discrimination 0.41 | not recorded | acute coronary syndrome 26%      |
| 5  | productive cough      | rule out acute_coronary_syndrome (26%); discrimination 0.38 | not recorded | acute coronary syndrome 26%      |
| 6  | pleuritic pain        | rule out acute_coronary_syndrome (26%); discrimination 0.29 | present      | acute coronary syndrome 26%      |
| 7  | fever                 | rule out acute_coronary_syndrome (8%); discrimination 0.33  | present      | community acquired pneumonia 33% |
| 8  | crackles              | expected entropy reduction 0.210 nats at cost 1.00          | present      | community acquired pneumonia 60% |
| 9  | ecg st changes        | required by guideline workup: Pericarditis assessment       | not recorded | community acquired pneumonia 68% |
| 10 | ecg pr depression     | required by guideline workup: Pericarditis assessment       | not recorded | community acquired pneumonia 68% |
| 11 | pericardial effusion  | required by guideline workup: Pericarditis assessment       | not recorded | community acquired pneumonia 68% |

It **committed** on community acquired pneumonia, which is the right answer. Note turn one: it asks for a D-dimer *to rule pulmonary embolism out*, not because embolism is likely — the red-flag weight makes exclusion worth paying for. Note also how many answers come back *not recorded*: a case report is not a questionnaire, and the loop has to reach a verdict on what the record happens to contain.

### 3 The knowledge base, and the number that matters

108

likelihoods invented

39%

carrying a citation

8/8

disease priors sourced

50

other invented parameters

**108 of 177 likelihoods are invented.** That is the most important fact about this project, and it is stated on its own line because burying it would be the error the whole design is organised against. Every number carries a tier, and the count is reported on every run.

#### MEASURED

A frequency counted in a named cohort or dataset. The only tier that supports a claim about the world rather than about a text.

#### NARRATIVE

A phrase from a reference text through a *fixed* rubric: “always”→0.95, “often”→0.55. Never re-tuned to produce a preferred answer.

#### INVENTED

A plausible guess by a non-clinician. Useful for exercising code paths and for nothing else.

#### Where the sourced numbers came from

| Source                    | Cells | What it is                                                                         |
|---------------------------|-------|------------------------------------------------------------------------------------|
| PIOPED-II-2007            | 16    | PIOPED II clinical tables — the arm investigated for embolism who did not have one |
| MSD-19E                   | 13    | Merck Manual 19e, narrative phrases through the fixed rubric                       |
| DDXPLUS                   | 9     | Frequencies counted in a simulator of ~1.3M patients                               |
| WANG-2005                 | 6     | JAMA review of dyspnoea in the ED, recovered by inverting likelihood ratios        |
| CAP-DIAGNOSTIC-MODEL-2024 | 5     | 954 acutely admitted patients, 265 with confirmed pneumonia                        |
| PIOPED-II-CTA-2006        | 5     | PIOPED II accuracy paper — CT angiography sensitivity and specificity              |
| MINIATI-2012              | 4     | 360 confirmed pulmonary embolism patients, Firenze                                 |
| STATPEARLS-PERICARDITIS   | 4     | StatPearls, acute pericarditis                                                     |
| DDIMER-TURBIDIMETRIC-META | 1     | D-dimer meta-analysis, 9 studies, n=1901                                           |
| TROPONIN-APE-2019         | 1     | 220 consecutive acute pulmonary embolism admissions                                |
| NTPROBNP-APE-2012         | 1     | 63 consecutive emergency embolism patients                                         |
| ZEGRE-HEMSEY-2018         | 1     | Acute coronary syndrome cohort                                                     |
| DDIMER-AECOPD-2013        | 1     | 148 COPD exacerbations investigated for embolism                                   |
| NOORAIN-2016              | 1     | COPD exacerbation troponin cohort                                                  |
| AECOPD-IMAGING-SR-2020    | 1     | Systematic review of imaging at COPD exacerbation                                  |

Both bulk routes are **proven** exhausted rather than assumed so. DDXPlus was closed by computing all 80 pairs its mapping can reach: 15 already sourced, and 13 unfilled pairs reading *exactly* 0.000 — the simulator omitting a finding from its rule base, not a frequency. Writing them would have asserted fever in 0% of pericarditis patients. Merck was closed by two complete passes over all eight relevant chapters.

# 4 What sourcing found, including what it took back

Sourcing a number is not the same as improving a model, and this project has the record to prove it. Every correction below replaced a guess with a measurement. Several made the results worse, and those are kept.

| Cell                         | Was   | Now  | What it means                                                                                                                                                                                                                      |
|------------------------------|-------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P(rest dyspnoea   pneumonia) | 0.05  | 0.67 | A faithful transcription of a Merck sentence about pneumonia at <i>all</i> severities, applied to an ED population who came in <i>because</i> they were breathless. It was arguing against the correct diagnosis in real patients. |
| P(raised BNP   embolism)     | 0.20  | 0.62 | Asserted the opposite of the mechanism: embolism strains the right ventricle, which is what releases the peptide.                                                                                                                  |
| P(troponin   embolism)       | 0.25  | 0.53 | Sat <i>below</i> the sourced 0.32 for COPD — claiming a COPD flare raises troponin more often than a pulmonary embolism does.                                                                                                      |
| P(CTPA defect   embolism)    | 0.95  | 0.83 | A safety correction. At 0.95 a negative scan read as near-conclusive; the measured sensitivity means roughly <b>one embolism in six is missed</b> .                                                                                |
| P(immobility   embolism)     | 0.61  | 0.25 | A simulator value against 192 real patients. The same gap as pleuritic pain, 71% simulated against 33% observed.                                                                                                                   |
| P(leg swelling   oedema)     | 0.999 | 0.50 | DDXPlus recorded it in 7202 of 7205 simulated patients. A likelihood of 0.999 should have been suspicious on sight; it survived because it carried a citation.                                                                     |

## A claim withdrawn, not just a number changed

The scope document once recorded, as a finding, that “*exertional chest pain no longer raises acute coronary syndrome*” — DDXPlus scored it at 0.77 in pulmonary oedema against 0.36 in ACS, and that was written up as something the sourced numbers had revealed.

*It was a mapping error. The DDXPlus question behind it asks “Do you have **symptoms** that are increased with physical exertion but alleviated with rest?” — symptoms, not chest pain. Heart-failure patients answer yes because their breathlessness does exactly that. The mapping was withdrawn and coverage fell.*

The mapping file's own docstring, written before the mistake was made, predicts it precisely: a mis-mapped number is *worse* than the invented one it replaced, because it arrives with 1.3M patients behind it. Three of the ten original mappings have now failed the same way: the comment beside each paraphrased the question instead of quoting it.

## The rubric that turns a sentence into a number

A reference text says a finding is “usually” present. Turning that into 0.75 is a judgement, and the only way to stop it being made afresh each time — conveniently, in whichever direction helps — is to fix the map once and apply it blindly.

| Phrase in the source text         | Value | Range     | Phrase in the source text | Value | Range     |
|-----------------------------------|-------|-----------|---------------------------|-------|-----------|
| always                            | 0.95  | 0.90–0.99 | less common               | 0.20  | 0.08–0.40 |
| characteristic / hallmark         | 0.85  | 0.70–0.95 | occasional                | 0.20  | 0.08–0.35 |
| usually                           | 0.75  | 0.60–0.90 | uncommon                  | 0.08  | 0.02–0.20 |
| common / frequent                 | 0.60  | 0.40–0.80 | rare / not typical        | 0.05  | 0.01–0.15 |
| often                             | 0.55  | 0.35–0.75 | absent                    | 0.02  | 0.01–0.08 |
| can occur / may occur / sometimes | 0.30  | 0.15–0.50 |                           |       |           |

13 of the 69 sourced cells came through this map, which has never been re-tuned to change an outcome.

# 5 What checks the numbers, and what checks the documents

Every test in a normal project checks outputs. The knowledge base is inputs, and for most of this project's life nothing looked at it at all. Three audits now do.

**Ordering audit — 23 claims, 86 comparisons**

For each finding: which disease must lead the column, over which rivals, and why. Reasons are textbook-level so a clinician can accept or reject each in one sentence. They constrain *ordering*, not magnitude — ordering is what a non-clinician can assert honestly.

**2 comparisons currently fail**, listed with the reason each is tolerated.

**50 parameters nobody was counting**

The provenance report covered likelihoods and priors. It did not count 6 correlation weights, 29 acquisition costs, 5 gate thresholds, 3 loop budget, 7 selector constants. The scope document already said why that is dangerous, about the priors: *an invented number the audit cannot name reads as an absence of a problem*. The hole was closed there and nobody asked whether it existed elsewhere.

**Documentation drift guard**

Every coverage figure quoted in SCOPE, DESIGN, RESPONSIBLE\_AI and the write-up is checked against a live run. Prose drifts silently because nothing executes it. The guard once drifted itself — it hardcoded a denominator that stopped matching any document, and passed. It now reads the denominator from the knowledge base.

**7 concepts given a threshold**

Every concept had an HPO label naming a qualitative state — “Hypoxemia”, “Elevated circulating D-dimer” — and no cutoff, while every study reports one. That is *why* three columns were unsourceable: matching a source to a concept needed a judgement call every time.

## What an ordering claim looks like

| Finding                      | Must lead                    | Over | Because                                                          |
|------------------------------|------------------------------|------|------------------------------------------------------------------|
| lab:raised_troponin          | acute coronary syndrome      | 6    | myocardial injury is what defines the diagnosis                  |
| lab:raised_troponin          | pulmonary embolism           | 3    | right ventricular strain releases troponin; a pneumonia does not |
| imaging:ctpa_filling_defect  | pulmonary embolism           | 4    | the finding names the diagnosis                                  |
| imaging:cxr_consolidation    | community acquired pneumonia | 7    | consolidation is the radiological definition of pneumonia        |
| imaging:cxr_pulmonary_oedema | acute pulmonary oedema       | 4    | the finding names the diagnosis                                  |
| lab:raised_bnp               | acute pulmonary oedema       | 5    | ventricular stretch is the mechanism that releases it            |
| lab:raised_d_dimer           | pulmonary embolism           | 6    | the one thrombotic diagnosis in the differential                 |

Seven of 23. Each reason is short on purpose: a clinician reading it can accept or reject the claim without reading any code, and rejecting one invalidates a specific set of cells rather than the whole table.

## What the audit caught that the tests did not

Five errors, none found by a failing test, and four of them in pulmonary embolism — in a system built around not missing it. The technique costs nothing: sort each column, ask whether the ordering matches what the diseases do, and look hardest where a sourced value sits below an invented one. Two of the five are in the table in section 4; the natriuretic-peptide and troponin inversions are the others, and both were *sourced* cells sitting below invented rivals.

# 6 How it is evaluated, and which numbers to distrust

## Three case sets, doing three different jobs

| Set           | n  | What it is for                                                                                                                                                                                                                     | Result                    |
|---------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Fixtures      | 10 | Hand-written during development. <b>Saturated</b> — the system sees them however the mechanisms are set, so they can no longer separate a good change from a bad one.                                                              | 8/10 top-1                |
| Hard cases    | 5  | Five named diagnostic traps written to replace them: cardiac asthma, myopericarditis, silent ischaemia, pneumonia in a COPD patient, asthma in a never-smoker. Written from the clinical picture <i>before</i> the model saw them. | 5/5 top-1, 0 wrong        |
| Real patients | 19 | Hand-extracted from open-access PMC case reports. Not invented, not generated, never seen by the knowledge base.                                                                                                                   | 4 correct, <b>0 wrong</b> |

## The required baselines, and the column that makes them fair

| System                        | top-1 | top-5  | MRR   | evidence seen |
|-------------------------------|-------|--------|-------|---------------|
| retrieval-only (no inference) | 57.1% | 100.0% | 0.786 | 27.0          |
| single-pass (KB, no loop)     | 85.7% | 100.0% | 0.929 | 27.0          |
| agentic loop                  | 71.4% | 100.0% | 0.833 | 11.1          |

Both baselines are handed the **complete** record — all 27.0 findings. The loop must ask for what it wants and pay per question, and here reaches its answer on 11.1 of them, 41% of the evidence. **Read the accuracy column and the loop loses.** That is reported rather than smoothed away, and so is the second uncomfortable reading: a retrieval-only baseline with no inference at all lands close to the loop, which is a finding about how easy this benchmark is, not about how good the system is.

## Calibration and selective prediction

|                                            |                             |                                         |                                        |
|--------------------------------------------|-----------------------------|-----------------------------------------|----------------------------------------|
| <b>0.286</b><br>expected calibration error | <b>0.170</b><br>Brier score | <b>43%</b><br>coverage (cases answered) | <b>0.127</b><br>AURC (lower is better) |
|--------------------------------------------|-----------------------------|-----------------------------------------|----------------------------------------|

Mean confidence is 67.2% against 71.4% accuracy — on this split the system is *underconfident* by 4.3%, not over. The gate answers 43% of cases and is right on 100% of those, against 71% if it were forced to answer everything. That gap is the entire claim: the value is in what it declines. It is also 7 cases, so the calibration figures above have almost no resolution — three bins would each hold two cases.

## The numbers that are not results

|                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>20/20 on synthetic cases</b><br>Worse than uninformative. Those cases were generated <i>from the knowledge base the agent reasons over</i> , so the score measures self-consistency. Every generated id carries a <i>synthetic-</i> prefix so one reaching a metrics table is visible. | <b>Agreement with DDXPlus</b><br>Bounded by what DDXPlus is: a rule-based simulator, which a naive-Bayes model recovers partly by construction. Where a real cohort disagreed — pleuritic pain, 71% simulated against 33% observed — the disagreement is reported rather than reconciled. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7 The 19 real patients

The only evidence here that comes from actual people: ten in a first pass, eight more in a second under a rule fixed before any full text was read. Still far too few to support an accuracy claim, and reported this small on purpose rather than not reported at all.

| Case                                                                                                                              | True diagnosis               | Verdict                     | Why it stopped                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------|---------------------------------------------------------------------|
| 78-year-old man, seven days of progressive shortness of breath on exertion, no chest pain                                         | Pulmonary embolism           | Escalated at 41% — ranked 2 | time-critical diagnosis not excluded: pulmonary_embolism (37%)      |
| 64-year-old woman, 12 hours of chest pain and shortness of breath, worse lying down and on deep breathing, better sitting forward | Pericarditis                 | Escalated at 27% — ranked 4 | top-1 confidence 27% below threshold 65%                            |
| 32-year-old man, burning pleuritic chest pain radiating to the left shoulder with shortness of breath, fever the day before       | Acute coronary syndrome      | Escalated at 41% — ranked 4 | top-1 confidence 41% below threshold 65%                            |
| 19-year-old woman, sudden trouble breathing and chest pain with facial numbness after becoming anxious during an exam             | Panic attack                 | Escalated at 25% — ranked 2 | top-1 confidence 25% below threshold 65%                            |
| 23-year-old man, one day of right-sided pleuritic chest pain, haemoptysis and fever                                               | Community-acquired pneumonia | Committed — correct         |                                                                     |
| 45-year-old woman, two weeks of worsening shortness of breath with orthopnoea, palpitations and massive leg swelling              | Acute pulmonary oedema       | Committed — correct         |                                                                     |
| 56-year-old man, 20 days of worsening dyspnoea worse in the mornings, with productive cough and fever                             | COPD exacerbation            | Escalated at 52% — ranked 2 | top-1 confidence 52% below threshold 65%                            |
| woman in her mid-30s, two days of shortness of breath, nonproductive cough, congestion and rhinorrhoea                            | Asthma exacerbation          | Escalated at 29% — ranked 6 | time-critical diagnosis not excluded: acute_coronary_syndrome (26%) |
| 36-year-old man, one day of fever, cough and chest pain with sudden breathlessness                                                | Pulmonary embolism           | Escalated at 21% — ranked 3 | time-critical diagnosis not excluded: pulmonary_embolism (18%)      |
| 85-year-old woman, five days of cough with blood-streaked sputum and breathlessness                                               | Community-acquired pneumonia | Escalated at 28% — ranked 2 | top-1 confidence 28% below threshold 65%                            |
| 33-year-old man, two days of mild pleuritic chest pain and a dry cough, ten days after a transcontinental flight                  | Pulmonary embolism           | Escalated at 34% — ranked 3 | time-critical diagnosis not excluded: pulmonary_embolism (15%)      |
| 53-year-old man, six hours of right subscapular pain followed by dyspnoea, fever, cough and haemoptysis                           | Community-acquired pneumonia | Committed — correct         |                                                                     |
| 83-year-old man, sudden nocturnal dyspnoea with cough and sweating that woke him from sleep, no chest discomfort                  | Acute coronary syndrome      | Escalated at 61% — ranked 1 | time-critical diagnosis not excluded: pulmonary_embolism (15%)      |

| Case                                                                                                                                                   | True diagnosis         | Verdict                     | Why it stopped                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------|---------------------------------------------------------------------|
| 44-year-old man, four weeks of worsening exertional breathlessness, paroxysmal nocturnal dyspnoea and orthopnoea, admitted with acute pulmonary oedema | Acute pulmonary oedema | Escalated at 65% — ranked 1 | top-1 confidence 65% below threshold 65%                            |
| 49-year-old woman with COPD, two weeks of worsening severe shortness of breath and productive cough, no chest pain                                     | COPD exacerbation      | Escalated at 48% — ranked 1 | top-1 confidence 48% below threshold 65%                            |
| 52-year-old man with childhood asthma, progressive dyspnoea requiring oxygen                                                                           | Asthma exacerbation    | Escalated at 31% — ranked 6 | time-critical diagnosis not excluded: acute_coronary_syndrome (26%) |
| 23-year-old woman, one week of worsening sharp pleuritic chest pain radiating to the back, with dyspnoea                                               | Pericarditis           | Committed — correct         |                                                                     |
| 21-year-old woman with a history of panic attacks, chest tightness, shortness of breath and tingling of the lips and fingertips                        | Panic attack           | Escalated at 35% — ranked 4 | top-1 confidence 35% below threshold 65%                            |
| 23-year-old man, acute precordial chest pain radiating to the neck and back, worse leaning forward, with fever and a racing heart                      | Pericarditis           | Escalated at 60% — ranked 1 | top-1 confidence 60% below threshold 65%                            |

**0 wrong commits — that is the number to look at.** Calibrated abstention is the claim, so the count that matters is not how many it answered (4 of 19) but how many it answered wrongly. For the first ten patients that number was zero, and the documents said so. The second pass took it away, and getting it back cost three separate fixes and a caveat.

*A 23-year-old woman, a week of pleuritic chest pain, tachycardic, white count 13.0, a “pulmonary infiltrate” on chest imaging, CT negative for embolism. The model committed to pneumonia at 78%. She had pericarditis, confirmed on tissue after a large pericardial effusion was drained through a surgical window. The knowledge base has no concept for a pericardial effusion, cardiomegaly or PR depression — the three findings that decide the case — so pneumonia explained everything it could see, and the gate’s sixth condition, which asks whether the findings are explained by something, was satisfied. Section 8 describes this failure in the abstract. This was it on a real patient, and the rule was to keep whatever came.*

Nothing about the extraction is strained: “infiltrate” was mapped to consolidation the same way the first pass mapped “basal infiltrate”, and the report says what it says. The first-pass claim of zero wrong commits was a statement about ten patients whose deciding findings happened to be in the vocabulary. One patient whose deciding finding is not was enough to end it.

## What it took to get that case right, and why to distrust it

### 1 Vocabulary

Effusion and PR depression added as concepts, sourced for pericarditis, with 21 invented rival cells because a concept only one disease lists is inert. Coverage *fell* to 38%. With both findings visible, pericarditis rose from 2% to 15%. Not enough, and the loop never asked for either.

### 2 A weighting defect

Findings asked for and never recorded were counting as correlated partners and discounting the ones that were observed. Fixed: real-case mean rank 3.22 to 2.50, one more correct commit — and this case went to **93%** pneumonia, because the wrong answer's evidence stopped being discounted too.

### 3 A workup the mechanism lacked

ESC 2015: ECG and echocardiography are Class I in suspected pericarditis. A third presentation-triggered workup, armed by pleuritic pain, asks for both. The case now commits correctly at 92%.

Written after seeing the case it fixes. It transcribes a guideline, reads the presentation and never the posterior, and moves no other set. Tested on the next pericarditis chosen by rule: it fired as designed and changed nothing — escalated at 60%, truth first, either way.

## How the 19 were chosen

### The extraction protocol

Each case comes from an open-access PMC case report with a confirmed final diagnosis. Findings are transcribed as present, absent, or *not recorded* — the third value matters most, because it is the one a generated corpus never produces.

Only what the source states explicitly becomes a True or False. Silence stays silence, because reading it as a negative would manufacture data the report never gave.

### First pass: ten

Four picked to be genuinely hard, four for the opposite reason — clear-cut, explicit vitals — and two more for the commonest causes on epidemiology, under a rule fixed *before* any was run.

Two candidates rejected, both written down.

### Second pass: eight, one per diagnosis

One title-restricted PMC query per condition, candidates taken in **order**, each accepted or rejected on four stated criteria, every rejection logged with its letter. All eight were added before any touched the model.

27 rejections, mostly mimics and in-patient events that were never a presentation. Result, under the vocabulary of the time: 0 correct, 1 wrong, 7 escalations with the truth ranked first in four.

Of 15 escalations, 6 stop because a time-critical diagnosis is still above the red-flag tolerance and 9 because the top hypothesis is stuck between about 27% and 65% against a 65% floor with nothing informative left to ask. That last column used to read differently. The packet appends a note when a workup item was sought and not recorded — *D-dimer, not available* on most of these — and the engines once put the blocker last, so the note was what a reader saw first and took for the cause. It was not: pulmonary embolism sat at 2–5% on every one of them, and an earlier draft of this page, and the author, proposed a fix for a problem that was not there. The order is now fixed where the reason is made. The real blocker is a thin record meeting a knowledge base that cannot separate pneumonia, pulmonary oedema and COPD on the findings a case report happens to give.

## What an escalation actually hands back

Case pmc-5841117, the pericarditis. Not a shrug — a ranked differential, the reason it stopped, and the cited evidence behind each line.

| Rank | Hypothesis                   | P   | Evidence the ranking rests on                                                                                                      |
|------|------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 1    | acute coronary syndrome      | 27% | lab:raised_troponin present; $P(\text{lab:raised\_troponin} \mid \text{acute\_coronary\_syndrome}) = 0.90$ ; likelihood ratio 2.28 |
| 2    | community acquired pneumonia | 25% | pleuritic_pain present; $P(\text{pleuritic\_pain} \mid \text{community\_acquired\_pneumonia}) = 0.53$ ; likelihood ratio 1.70      |
| 3    | acute pulmonary oedema       | 25% | no positive finding in the record                                                                                                  |
| 4    | pericarditis                 | 17% | pleuritic_pain present; $P(\text{pleuritic\_pain} \mid \text{pericarditis}) = 0.69$ ; likelihood ratio 2.20                        |

**Why it stopped:** *top-1 confidence 27% below threshold 65%; no remaining action would meaningfully narrow the differential; Pericarditis assessment could not be completed -- sought but not available: exam:ecg\_pr\_depression, imaging:pericardial\_effusion.* The true diagnosis does not lead here, which is a real miss and is counted as one — but the system hands back a ranked list and a reason instead of committing to the hypothesis at the top.

# 8 What is wrong with it

## One standing red mark

P(hypoxia | acute pulmonary oedema) is an invented 0.40, below COPD and pulmonary embolism, when alveolar flooding is the mechanism that *defines* the disease.

Unsourceable for a structural reason: saturation is part of how the severe presentation is identified. Trials enrol on it; the AHEAD registry uses it in the classification criterion itself. Taking that would be circular.

## Five D-dimer rivals understated

Three independent lines of evidence say they are about twice too low. The sources either contradict each other or need a distributional assumption no other value here depends on.

Recorded as a measured defect rather than papered over with a number nobody chose.

## Confident errors the gate cannot catch

Condition 6 asks whether the findings are explained by *something*. In a masquerade they are. It happened on a real patient: a pericarditis committed as pneumonia at 78%, then 93%. What finally caught it was not a threshold but a presentation-triggered workup that asks for the echo a myopic selector never would, because pericarditis sat at 4% when the loop decided. **The gate still cannot see a masquerade; it can only be made to look.**

## Calibration is not fitted at all

The scaler refuses below 30 labelled cases, correctly. The calibration split is three, so the temperature is the untouched default.

The report used to print "temperature (fitted) 1.00", which reads as a fit that found the posterior already calibrated. It now says no fit happened.

## The failing comparisons, printed rather than described

| Finding      | Should lead            | P    | But this rival has | P    |
|--------------|------------------------|------|--------------------|------|
| exam:hypoxia | acute pulmonary oedema | 0.40 | copd exacerbation  | 0.55 |
| exam:hypoxia | acute pulmonary oedema | 0.40 | pulmonary embolism | 0.60 |

That is the whole list — 2 of 86 comparisons. The audit script exits non-zero on any failure that is not on the tolerated list, so a new inversion cannot enter quietly.

## Not delivered, and it will not be from here

| Deliverable                      | Why not                                                                                                                                                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clinician-graded evidence chains | <b>No collaborating clinician was available.</b> Published decision rules substitute for which findings support which cause, and likelihood ratios for strength. Grading the chain has <b>no substitute and is not delivered.</b> |
| The mandated model, run live     | No API credit for the project's duration. The LLM layer has been a swappable Protocol since week 1 for exactly this reason, and that default has never been called.                                                               |
| AgentClinic as the harness       | Needs its own API key plus a fork to accept this agent in place of its own. Not a configuration change, and not attempted.                                                                                                        |
| UMLS relations                   | Probed and found unusable: overwhelmingly translations, ICD crosswalks and MedDRA groupings, with one clinical label that mixes symptoms, risk factors and treatment complications without marking which is which.                |

## What would have to happen before any of this is usable

In order, and none of it is engineering: a clinician reads the 23 ordering claims and rejects the ones that are wrong; the remaining 108 invented likelihoods are sourced or removed; thirty or more labelled cases are assembled so the temperature can actually be fitted; and the real-case set grows past the point where a two-case swing is noise. The

code is ready for all four. None of them is a code change.

# 9 What this is, in one page

A working research prototype whose engineering is complete and whose clinical content is not validated. That is the honest one-line description and everything above is an elaboration of it.

## What would survive a hostile read

**It knows what it does not know**

108 of 177 likelihoods invented, 50 further parameters invented, reported on every run, with a regression test that fails when the documents drift from the live figure.

**It refuses rather than guesses**

0 wrong commits across 19 real patients and 5 deliberately hard ones, against 15 escalations that each name what was sought and not obtained. The one wrong commit there was is written up, with the three fixes it took and why to distrust the last of them.

**It records the conclusions it took back**

A correlation measurement that reversed eight times before anyone recognised the instrument was spent — and the one that never had, three wrong commits against none on real patients, which a pneumonia sourcing pass took to none against none. A published “finding” that was a mapping bug. A claim of zero wrong commits that lasted until the eighteenth patient.

**It checks its own inputs**

23 ordering claims over 86 comparisons, 52 direction claims from the scope document, 2 distribution-free bounds from cohort quartiles, and 7 concepts given the operational threshold that makes sourcing possible at all.

*Three separate times, sourcing a number overturned something the author believed; twice it made a result worse, and those results were kept and written down. A system that cannot tell you which of its beliefs are invented cannot be corrected, and a project that only records the corrections that flattered it has not really recorded anything.*

## Running it

All of it runs from a clean checkout; the scripts as `python3 scripts/<name>`.

| Command                            | What it gives you                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>pytest tests/ -q</code>      | 170 tests, including the two-engine equivalence check and the documentation drift guard                            |
| <code>run_eval.py</code>           | The full metrics report and the two required baselines, with the evidence column                                   |
| <code>eval_real_cases.py</code>    | The 19 real patients, one verdict at a time, with the reason for each escalation                                   |
| <code>plausibility_check.py</code> | The knowledge-base audits: 23 ordering claims over 86 comparisons, and 52 direction claims from the scope document |
| <code>consult.py</code>            | An interactive consultation in the terminal, one question at a time                                                |
| <code>webui/server.py</code>       | The same loop on a web page, with a live differential and the provenance of every cited entry                      |

No install, no API key, fully offline and deterministic.

### A closing caution

Nothing here has been reviewed by a clinician, most of the knowledge base is invented, and the strongest evidence in the project is 19 patients. **This is not a diagnostic tool and must not be used as one.** What it is, is a system that will tell you exactly which parts of itself you should not trust — and that is the part worth keeping.